Neural Networks applied in NLP
Lab 3
2026-2027

Get familiar with syntactic and dependency parsing by generating the corresponding trees for the following sentences:

Visiting professors can be annoying.
Old men and women were waiting outside.
The reporter interviewed the actor with the camera near the entrance.

Tasks:
1. (0.15p) Write a suitable CFG grammar, able to correctly parse all sentences, with all their senses. Use as inspiration the grammars discussed in class.

2. (0,15p) Use existing modules to extract phrase structure trees for these sentences using your grammar. You may use, for example: nltk.ChartParser and/or nltk.EarleyChartParser.

3. (0.25p) Implement a dependency parser (using, for instance, spaCY, NLTK or Stanza) and parse the three sentences above.

4. (0,30p) Write a program which transforms a Context Free Grammar, as the one discussed in class,
into a Chomsky Normal Form grammar, also converting the probabilities. Test it on the grammar from the course.

5. (0,15p) Describe an application n which constituency and/or dependency parsing can provide useful information. Explain briefly why. (1/2 pag.)